KAIZEN MATHS

Lesson starter

Percentage Applications: Finance and Contexts

Scientific & Complex

Name _____

S1

A solution contains 14.5g of salt in 387.5g of water. What is the concentration? How much salt is needed for 805.7g of the same solution?

S2

In a study of 832 participants, 671 showed positive results. Calculate the success rate and the 94% confidence interval with margin of error ±3.8%.

S3

A measurement gives 79.16 units, but the true value is 83.70 units. Calculate the absolute error, relative error percentage, and accuracy percentage.

S4

A solution contains 10.4g of salt in 175.7g of water. What is the concentration? How much salt is needed for 973g of the same solution?

Show your working

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NUM-07B4-L4 · S

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Exit ticket

Percentage Applications: Finance and Contexts

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Name _____

EA1

In a study of 531 participants, 330 showed positive results. Calculate the success rate and the 96% confidence interval with margin of error ±3.4%.

EA2

A measurement gives 56.61 units, but the true value is 57.05 units. Calculate the absolute error, relative error percentage, and accuracy percentage.

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Exit ticket

Percentage Applications: Finance and Contexts

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EB1

A solution contains 9.8g of salt in 174.2g of water. What is the concentration? How much salt is needed for 857.8g of the same solution?

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Scientific & Complex

Name _____

EA1

In a study of 531 participants, 330 showed positive results. Calculate the success rate and the 96% confidence interval with margin of error ±3.4%.

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A measurement gives 56.61 units, but the true value is 57.05 units. Calculate the absolute error, relative error percentage, and accuracy percentage.

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EB1

A solution contains 9.8g of salt in 174.2g of water. What is the concentration? How much salt is needed for 857.8g of the same solution?

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Show your working

Teacher answers & working

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S1

A solution contains 14.5g of salt in 387.5g of water. What is the concentration? How much salt is needed for 805.7g of the same solution?

Answer:

Concentration: 3.61%, salt needed: 29.09g

Working:

Total solution = 14.5g + 387.5g = 402g

Concentration means solute ÷ total solution × 100.

Concentration = (14.5 ÷ 402) × 100 = 3.61%

Salt needed = 3.61% of 805.7g

Salt needed = 0.0361 × 805.7 = 29.09g

S2

In a study of 832 participants, 671 showed positive results. Calculate the success rate and the 94% confidence interval with margin of error ±3.8%.

Answer:

Success rate: 80.65%, interval: 76.85% - 84.45%

Working:

Success rate = (671 ÷ 832) × 100 = 80.65%

Lower bound = 80.65% - 3.8% = 76.85%

Upper bound = 80.65% + 3.8% = 84.45%

94% confidence interval: 76.85% - 84.45%

S3

A measurement gives 79.16 units, but the true value is 83.70 units. Calculate the absolute error, relative error percentage, and accuracy percentage.

Answer:

Absolute error: 4.54, relative error: 5.42%, accuracy: 94.58%

Working:

Absolute error = |79.16 - 83.70| = 4.54

Relative error % = (4.54 ÷ 83.70) × 100 = 5.42%

Accuracy = 100% - 5.42% = 94.58%

Teacher answers & working

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S4

A solution contains 10.4g of salt in 175.7g of water. What is the concentration? How much salt is needed for 973g of the same solution?

Answer:

Concentration: 5.59%, salt needed: 54.39g

Working:

Total solution = 10.4g + 175.7g = 186.g

Concentration means solute ÷ total solution × 100.

Concentration = (10.4 ÷ 186.1) × 100 = 5.59%

Salt needed = 5.59% of 973g

Salt needed = 0.0559 × 973 = 54.39g

EA1

In a study of 531 participants, 330 showed positive results. Calculate the success rate and the 96% confidence interval with margin of error ±3.4%.

Answer:

Success rate: 62.15%, interval: 58.75% - 65.55%

Working:

Success rate = (330 ÷ 531) × 100 = 62.15%

Lower bound = 62.15% - 3.4% = 58.75%

Upper bound = 62.15% + 3.4% = 65.55%

96% confidence interval: 58.75% - 65.55%

EA2

A measurement gives 56.61 units, but the true value is 57.05 units. Calculate the absolute error, relative error percentage, and accuracy percentage.

Answer:

Absolute error: 0.44, relative error: 0.77%, accuracy: 99.23%

Working:

Absolute error = |56.61 - 57.05| = 0.44

Relative error % = (0.44 ÷ 57.05) × 100 = 0.77%

Accuracy = 100% - 0.77% = 99.23%

Teacher answers & working

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EB1

A solution contains 9.8g of salt in 174.2g of water. What is the concentration? How much salt is needed for 857.8g of the same solution?

Answer:

Concentration: 5.33%, salt needed: 45.72g

Working:

Total solution = 9.8g + 174.2g = 184g

Concentration means solute ÷ total solution × 100.

Concentration = (9.8 ÷ 184) × 100 = 5.33%

Salt needed = 5.33% of 857.8g

Salt needed = 0.0533 × 857.8 = 45.72g